Design and Development of a Novel Haptic Device for Plucked Musical Instrument AR Simulation

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***Abstract*—Haptic devices have emerged as a promising technology for enhancing the experience of playing virtual musical instruments by providing tactile feedback to the user. This paper presents a novel haptic device that emulates plucked musical instruments as part of an augmented reality (AR) system. This wearable device features a pair of parallel 5-bar mechanism that provides haptic feedback. While initially developed to simulate harp playing, this technology could also be applied to emulate fingerstyle guitars, banjos, and other plucked instruments. The system can detect the position of a finger relative to a virtual string's projected position and provide haptic feedback by moving a string piece against the fingertip. Upon detecting a plucking motion, the device plays a musical note based on the virtual string's projection the finger is closest to and moves the string piece back to its rest position. The device has been designed, developed, and tested to ensure functionality and user comfort.**

Keywords—Haptic Device; Five-bar Mechanism; Musical Instrument; Haptic Feedback; Forward Kinematics

I. Introduction

In various scenarios, musicians could benefit from the ability to practice or play their instruments, but are hindered by space limitations and the weight of the instrument(s). To address this, an augmented reality (AR) instrument would be highly advantageous. To fully replicate the experience of playing the instrument, a haptic interface would be necessary to simulate the sensation of the instrument responding to the user's actions.

There is much research in the literature on wearable haptic devices for different applications [1], [2], ranging from stiffness rendering to exoskeletons. Among these, only a few devices are designed for musical creation purposes. A glove equipped with vibrational haptic fingertip mechanisms that guide users on which notes to play on a virtual piano for practice purposes already exists [3].

Another study involves using a Phantom Premium 3.0 6-DoF interface to simulate playing a virtual six-stringed plucked instrument (a guqin) [4]. Although effective in simulating an instrument, the haptic feedback method used is not the same as playing an actual plucked instrument using finger pads or nails. A portable, wearable haptic feedback device would be more desirable for AR musical expression.

LinkTouch, while not designed for musical instrument emulation, is the closest existing device to the one designed in this paper. It is a 2-DoF system that uses a joint in a five-bar mechanism to create a motion across the finger pad [5]. In theory, this device could create a similar simulation to the one described in this project if some modifications were made to the angle of the mechanism in relation to the user's finger. However, it would not have the same level of realism as the current design in this paper. The current design uses a segment of a musical instrument string to simulate the feeling of accessing a full instrument string, resulting in the most authentic plucked string interaction achievable with a haptic device thus far.

This paper aims to design and implement an interface that can mimic the sensation of plucking strings on a virtual instrument, such as a guitar or harp. In this study, a novel haptic device is designed, developed, and evaluated successfully. Section II discusses the design of the device and the theory of the five-bar mechanism. In Section III, static force analysis of the mechanism has been presented. Section IV includes specifications of the hardware, controller unit and pluck detection and discusses the results obtain by the developed device. Finally, some concluding points are presented in Section V.

II. Design of the Haptic Device

A. Requirements

The main idea of the haptic device is to position an actual string piece in front of a fingertip. The position of the string piece must be actively controlled to enable the device to move it with respect to the user’s fingertip. A pluck detection technique is required to detect the plucking action of the user. In addition, a hand localization method needs to be implemented.

B. Initial Design

The initial design idea was developed, which involved placing a string component (as illustrated in Fig. 2) between two linked five-bar mechanisms. The arrangement allowed for two degrees of freedom (2DoF) of the string component, which could move towards and away from the user's finger as they moved it towards or away from a virtual string. To enable coupled motion, each five-bar mechanism was fitted with a single motor, and a bearing was attached across from the shaft of each motor to allow motion to propagate from each five-bar mechanism to the other.

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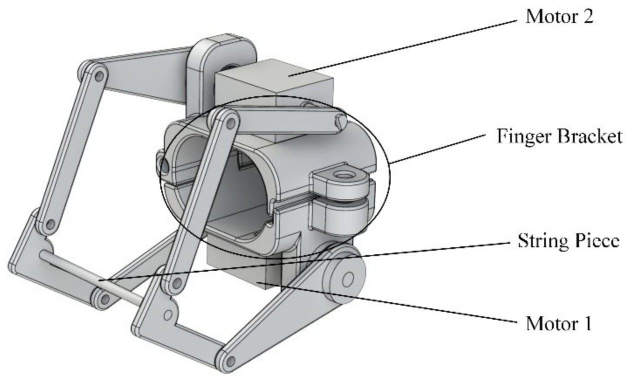

Fig. 1. Initial design of the haptic device.

C. Forward Kinematics of Five-bar Mechanism

As discussed, the initial design uses a pair of parallel five-bar linkage mechanisms. Since both five-bar linkages are assumed fully constrained to each other, only one of them is considered in calculations. The schematic of the five-bar mechanism is shown in Fig. 2(a) and with more details in Fig. 2(b). As can be seen in Fig. 2(b), the string piece is considered as the end-effector of the mechanism.

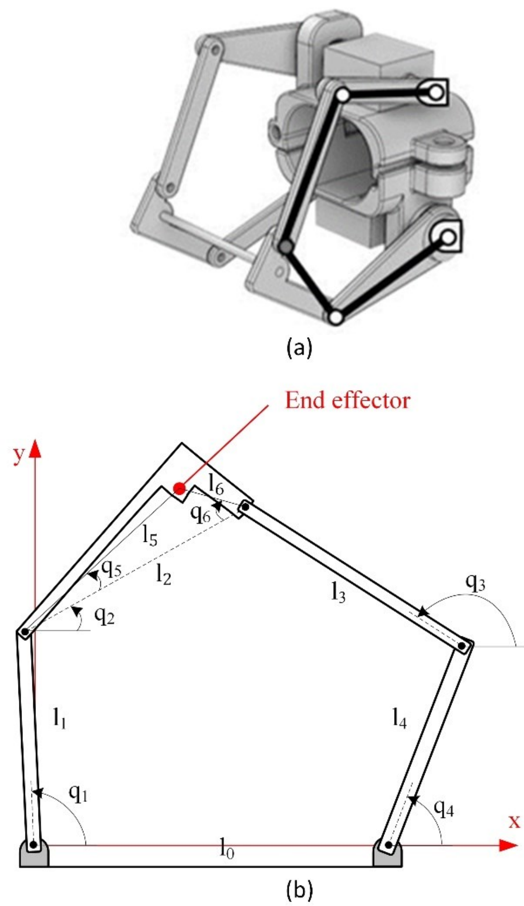

Fig. 2. (a) Schematic of the five-bar mechanism on the initial design; (b) Labeled schematic diagram of the mechanism.

Forward kinematics of the mechanism can be calculated by formulating loop closure equations based on Fig. 2(b) and decomposing real and imaginary components:

$$l_1 \cos(q_1) + l_2 \cos(q_2) = l_0 + l_4 \cos(q_4) + l_3 \cos(q_3) \tag{1}$$

$$l_1\, sin(q_1) + l_2\, sin(q_2) = l_4\, sin(q_4) + l_3\, sin(q_3) \tag{2}$$

Setting $c_i = cos(q_i)$, $s_i = sin(q_i)$ for the sake of brevity:

$$l_1 c_1 + l_2 c_2 = l_0 + l_4 c_4 + l_3 c_3 \tag{3}$$

$$l_1 s_1 + l_2 s_2 = l_4 s_4 + l_3 s_3 \tag{4}$$

The l_i values (link lengths) are fixed (known) and the dependent variables q_2 and q_3 could be determined for any given motor configuration, which governs the value of independent variables q_1 and q_4

$$tan^2\left(\frac{q_2}{2}\right)(-h_1 - h_3) + 2\, tan\left(\frac{q_2}{2}\right) h_2 + (h_1 - h_3) = 0 \tag{5}$$

where:

$$h_1 = l_1 l_2 c_1 - l_0 l_2 - l_2 l_4 c_4$$

$$h_2 = l_1 l_2 s_1 - l_2 l_4 s_4$$

$$h_3 = l_3^2 - l_0^2 - l_1^2 - l_2^2 - l_4^2 - l_0 l_4 c_4 + l_0 l_1 c_1 + l_1 l_4 c_1 c_4 + l_1 l_4 s_1 s_4$$

By solving the quadratic equation for q_2:

$$q_2 = 2tan^{-1}\left[\frac{-B \pm (B^2 - 4AC)^{\frac{1}{2}}}{2A}\right] \tag{6}$$

where:

$$A = -h_1 - h_3,$$
$$B = 2h_2,$$
$$C = h_1 - h_3$$

While two solutions will be produced, only one will match a configuration where the end effector appears in front of l_0 (i.e., with a positive y value). The other value may be safely ignored.

This same process can be done to eliminate q_2 and solve for q_3:

$$tan^2\left(\frac{q_3}{2}\right)(-h_4 - h_6) + 2\tan\left(\frac{q_3}{2}\right) h_5 + (h_4 - h_6) = 0 \tag{7}$$

where:

$$h_3 = l_2^2 - l_0^2 - l_1^2 - l_3^2 - l_4^2 - l_0 l_4 c_4 + l_0 l_1 c_1 + l_1 l_4 c_1 c_4 + l_1 l_4 s_1 s_4$$

$$h_4 = l_0 l_3 + l_3 l_4 c_4 - l_1 l_3 c_1$$

$$h_5 = l_3 l_4 s_4 - l_1 l_3 s_1$$

The solutions to q_3:

$$q_3 = 2tan^{-1}\left[\frac{-E \pm (E^2 - 4DF)^{\frac{1}{2}}}{2D}\right] \tag{8}$$

where:

$$D = -h_4 - h_6,$$
$$E = 2h_5,$$
$$F = h_4 - h_6$$

Again, two solutions will be found, but only one will produce a positive y value for the actual end effector. It is also possible to check the solutions of q_2 and q_3 by comparing the x and y end effector values that will be produced, eliminating any solutions that do not produce a match.

We can now find the x_{ee} and y_{ee} positions of the actual end effector:

$$x_{ee} = l_1 \cos(q_1) + l_5 \cos(q_2 + q_5) \tag{9}$$

$$y_{ee} = l_1 \sin(q_1) + l_5 \sin(q_2 + q_5) \tag{10}$$

III. Implementation of the Haptic Device

A. Hardware Specifications

Based on the structure shown in Fig. 2(b), the following dimensions are chosen for the manufactured prototype:

$l_0 = 3.2cm, l_1 = 2.2cm, l_2 = 2.35cm,$

$l_3 = 2.4cm, l_4 = 3cm, l_5 = 1.9cm, l_6 = 0.6cm,$

$q_5 = 10.8°, q_6 = 36.3°$

A prototype of the mechanism, composed of 3D printed PLA parts, various metal connectors, and two identical brushed DC motors with an embedded quadratic encoder for providing motion to the mechanism, is manufactured to be attached to the right forefinger of a user. The developed prototype with all the components is shown in Fig. 3.

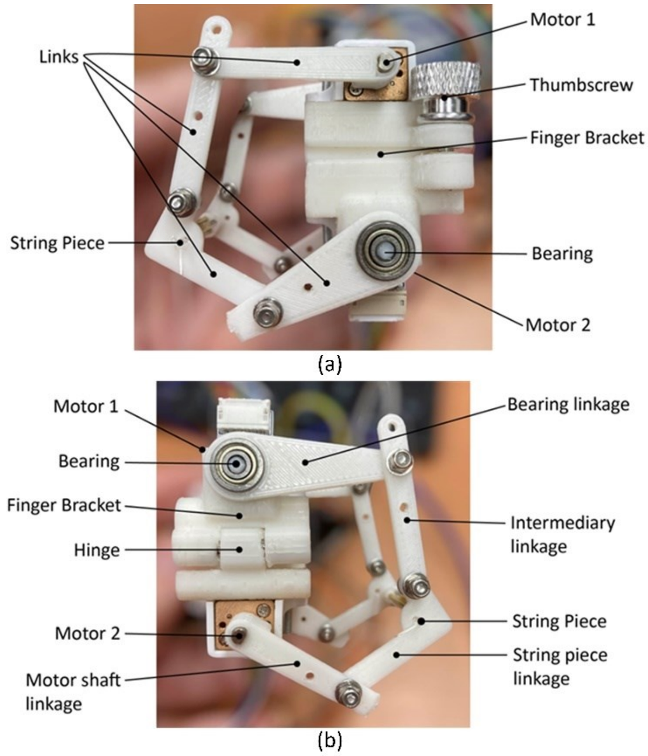

Fig. 3. Prototype of the haptic device from (a) right and (b) left views.

To close the mechanism around the user's finger, a hinge was created on one side of the finger bracket that holds the mechanism in place between the first and second joints of the finger. A thumbscrew was used to control the tightness of this closure. As a safety feature and a method of controlling the tension in the string piece, four spring-loaded shafts were added behind the string piece. In addition, the spring-loaded shafts act to synchronize the motions of the two parallel five-bar mechanisms.

B. Controller Unit

A motor driver and an Arduino control board are provided to be attached to the user's wrist. Two independent PID controllers are coded on the Arduino for motion control.

A pair of appropriate setpoints for each motor is hardcoded in the Arduino code (adjustable) and the motors are moved to "plucking" or "rest" configurations based on the output of the pluck detection system, shown in Fig. 4. The output of the pluck detection is transferred to the Arduino board using a serial connection.

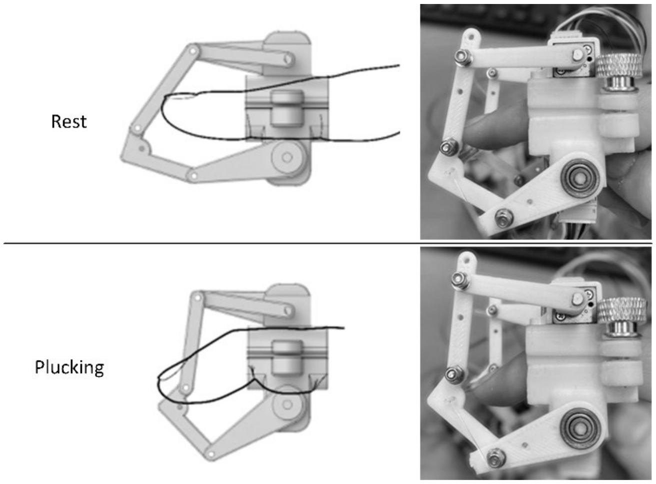

Fig. 4. "Rest" and "Plucking" configurations of the device.

C. Pluck Detection

A system for visual tracking of the user's hand and finger motion via a Leap Motion controller is provided. A Python program is developed to detect the extension/contraction of the index finger using the Leap Motion controller. Fig. 5 shows the output of the Leap Motion controller.

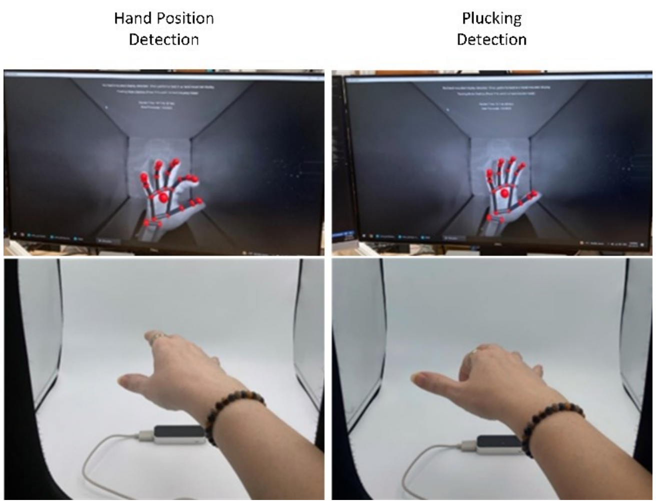

Fig. 5. Hand tracking and index finger extension/contraction detection using a Leap Motion controller.

As the user's hand moves within the motion detection area, the Leap Motion controller instructs the Arduino board to position the string component against the user's finger (i.e., "engaged" configuration). Once in position, if the user executes a plucking motion with their forefinger, the Leap Motion controller detects the motion and finger position and directs the Arduino board to move the string piece away from the finger. Meanwhile, the Leap Motion code plays one of three notes based on the finger's position within the range of motion capture. The concept of different strings being plucked to play different notes is reflected in the definition of three distinct motion capture zones. This process is shown schematically in Fig. 6.

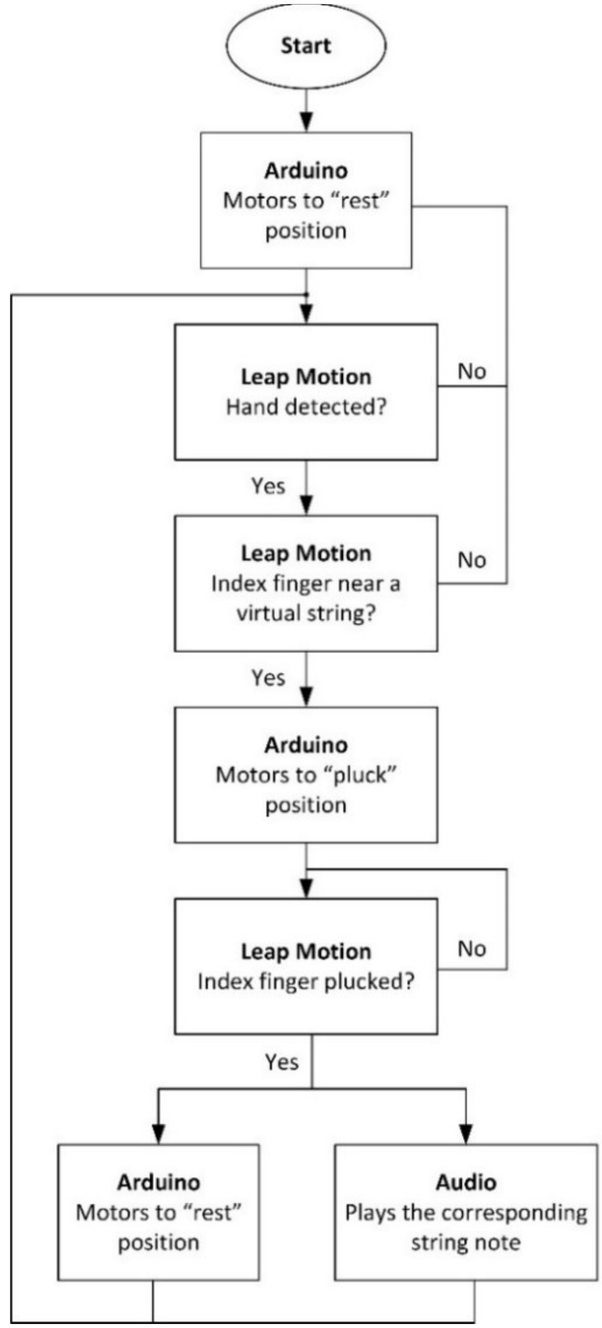

Fig. 6. Flowchart of the hand detection and finger plucking process.

IV. Results and Discussion

Given the equations of motion found in Section II, a MATLAB program was written to visualize the workspace for the end effector. The desired region of the workspace is shown in Fig. 7. By having proper setpoints for the motors, the string piece can be positioned at any point in the desired region of the workspace.

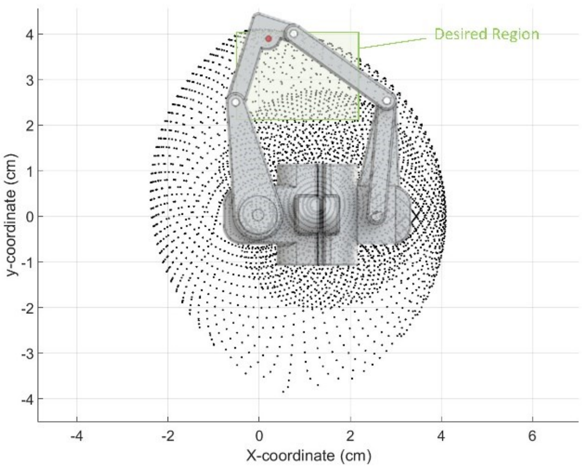

Fig. 7. Workspace of the five-bar mechanism and the desired region.

A wearable haptic feedback system with a parallel 5-bar mechanism was designed, developed, and tested for emulating plucked musical instruments in an augmented reality system. The mechanism, which fits on the forefinger of the user's right hand, detects the position of the user's finger relative to the projected position of a virtual string and provides haptic feedback by moving a string piece against the user's fingertip. Once the user plucks the string piece, the system sounds a MIDI note based on the projected virtual string closest to the finger's position and returns the string piece to a rest position. This process is shown in Fig. 8.

This design represents a breakthrough in the realm of haptic feedback musical instrument emulation, particularly for plucked musical strings. Its accuracy surpasses that of any existing haptic feedback musical instrument emulating device. Notably, extensive experiments have confirmed that the attachment of the haptic device does not interfere with the Leap Motion detection process, ensuring precise tracking of hand and finger positions. However, it is crucial to thoroughly investigate the potential impact of the device's weight on the overall user experience.

There are certain limitations when attempting to implement multiple devices simultaneously on different fingers. To overcome this, future iterations of the device should focus on a slimmer and more compact design, enabling the attachment of several devices on various fingers. It's worth noting that the current version of the device has been exclusively developed for plucking, and other actions, such as strumming, were not taken into consideration during the design stage.

V. Conclusions

In conclusion, the development of the haptic device has opened up new possibilities for enhancing the virtual musical instrument playing experience. The wearable haptic feedback device presented in this paper is a novel technology that emulates the plucking of musical instruments as part of an augmented reality system. Its two parallel five-bar mechanisms provide haptic feedback to the user, while its ability to detect the position of the finger and play a musical note based on the virtual string's projection adds to the immersive experience. The device has been designed, developed, and tested to ensure functionality and can be used to simulate the playing of various plucked instruments. With further advancements in haptics technology, this device has the potential to revolutionize the way musicians use and listeners experience virtual musical instruments. Further research is needed to address the weight-related user experience concerns and to refine the design to accommodate multiple devices across different fingers. By doing so, the device can cater to a broader range of musical techniques and deliver an even more immersive experience for musicians.

Acknowledgment

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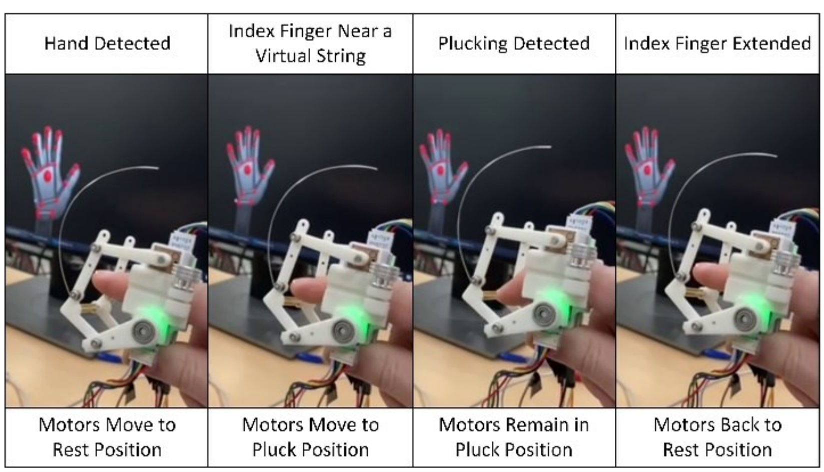

Fig. 8. Hand localization and plucking detection using a Leap Motion controller and movement of the haptic device from rest position to pluck position and vice versa based on the user's hand.

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